

RankBind (v5): how the four anti-shortcut mechanisms work

The shortcut to defeat: rank proteins by a protein-only prior (“this protein binds everything”), ignoring the ligand. Every score splits into three parts:

$s(L, P) =$

$\alpha(L)$

+

$\beta(P)$

+

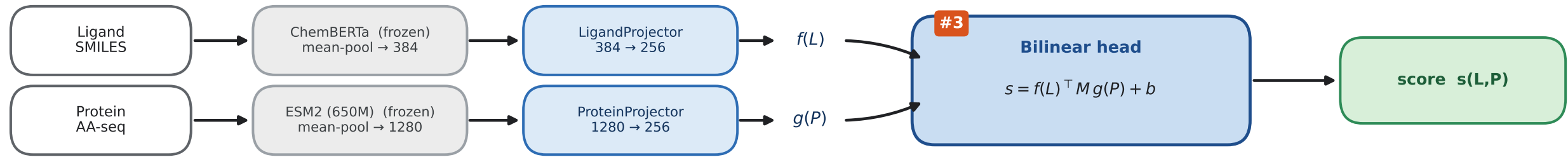
$\gamma(L, P)$

ligand-only

protein prior = the SHORTCUT

interaction = what we WANT

ARCHITECTURE (forward pass)



THE FOUR ANTI-SHORTCUT MECHANISMS — each drawn as its principle:

#1 Protein-balanced sampler

Each protein is drawn as a binder (+) and a non-binder (−) about equally often, so the per-protein positive-rate prior — what the null_prot_prior baseline exploits — drops to chance.

#2 Within-ligand margin loss

Fix the ligand L: P^+ must beat every P^- by a margin m .
L is shared $\Rightarrow \alpha(L)$ and bias b cancel in the difference, so the loss falls only by ranking proteins on $\gamma(L, P)$.

#3 Bilinear interaction head

MLP-concat head

can learn $s \approx \beta(P)$ (ignores the ligand)

Bilinear head (RankBind)

$+ b$
P enters ONLY via $\times$ with L; only the scalar b is ligand-free

No additive protein-only term (unlike MLP-concat), so the head cannot just add a protein prior. A structural bias, not a guarantee: the margin loss is the dominant driver; DeltaField makes it exact.

#4 Hard-negative mining

P^+ (true binder)
top-50 confusers $\rightarrow$ draw $k=4$ here (HARD)
random negatives (easy: margin already satisfied)

Each epoch, re-score all (positive-ligand $\times$ train-protein) pairs and draw the $k=4$ negatives from the current top-50 confusers, not random easy ones $\rightarrow$ the margin keeps biting.

Net effect (v4, 3-seed mean $\pm$ std):

matrix MRR 0.33 ± 0.07 · Hit@10 0.76 ± 0.10 · Gini-residual -0.21 ± 0.02 · Top-10 Jaccard vs null ≈ 0.00

Causal control: keep this architecture but swap the margin loss for BCE $\rightarrow$ the shortcut returns (global AUC 0.92, but matrix MRR ≈ 0.01 , Jaccard-vs-null 0.43). The loss, not the features, is what removes the prior.